

Week 1 Feedback Update Notes

1. Naming Requirement Confirmed

- Week1.pdf requires the analysis notebook name: EDA_final.ipynb.
- Week1.pdf requires the cleaned dataset name: cleaned_sales.csv.
- The updated package now includes both required deliverables.

2. Change Summary

- Reduced repeated DataFrame copies by using one boolean filter mask and chained processing.
- Read CustomerID as string during CSV import because it is an ID, not a numeric measure.
- Added EDA_final.ipynb and set cleaned_sales.csv as the main cleaned output file.

3. Before / After Code Changes

A. Memory Optimization

Before

```
df_clean = df_raw.copy()
df_clean = df_clean.dropna(...).copy()
df_clean = df_clean[(df_clean["Quantity"] > 0) & (df_clean["UnitPrice"] > 0)].copy()
df_clean = df_clean.drop_duplicates().copy()
```

After

```
valid_sales_mask = (
    df_clean[["Quantity", "UnitPrice", "CustomerID", "OrderDate"]].notna().all(axis=1)
    & (df_clean["Quantity"] > 0)
    & (df_clean["UnitPrice"] > 0)
)

df_clean = (
    df_clean.loc[valid_sales_mask]
    .drop_duplicates()
    .assign(Revenue=lambda x: x["Quantity"] * x["UnitPrice"])
)
```

B. CustomerID Handling

Before

```
df_raw = pd.read_csv(DATA_PATH, encoding="latin1")
df_clean["CustomerID"] = pd.to_numeric(df_clean["CustomerID"], errors="coerce")
df_clean["CustomerID"] = df_clean["CustomerID"].astype("int64").astype("string")
```

After

```
df_raw = pd.read_csv(
    DATA_PATH,
    encoding="latin1",
    dtype={"CustomerID": "string"},
)
CustomerID=df_raw["CustomerID"].str.strip().replace("", pd.NA)
```

C. Deliverable Naming

Before

```
Working notebook: 1.1_Preprocess.ipynb  
Earlier cleaned file: cleaned_data.csv
```

After

```
Required notebook: EDA_final.ipynb  
Required cleaned file: dataset/cleaned_sales.csv
```